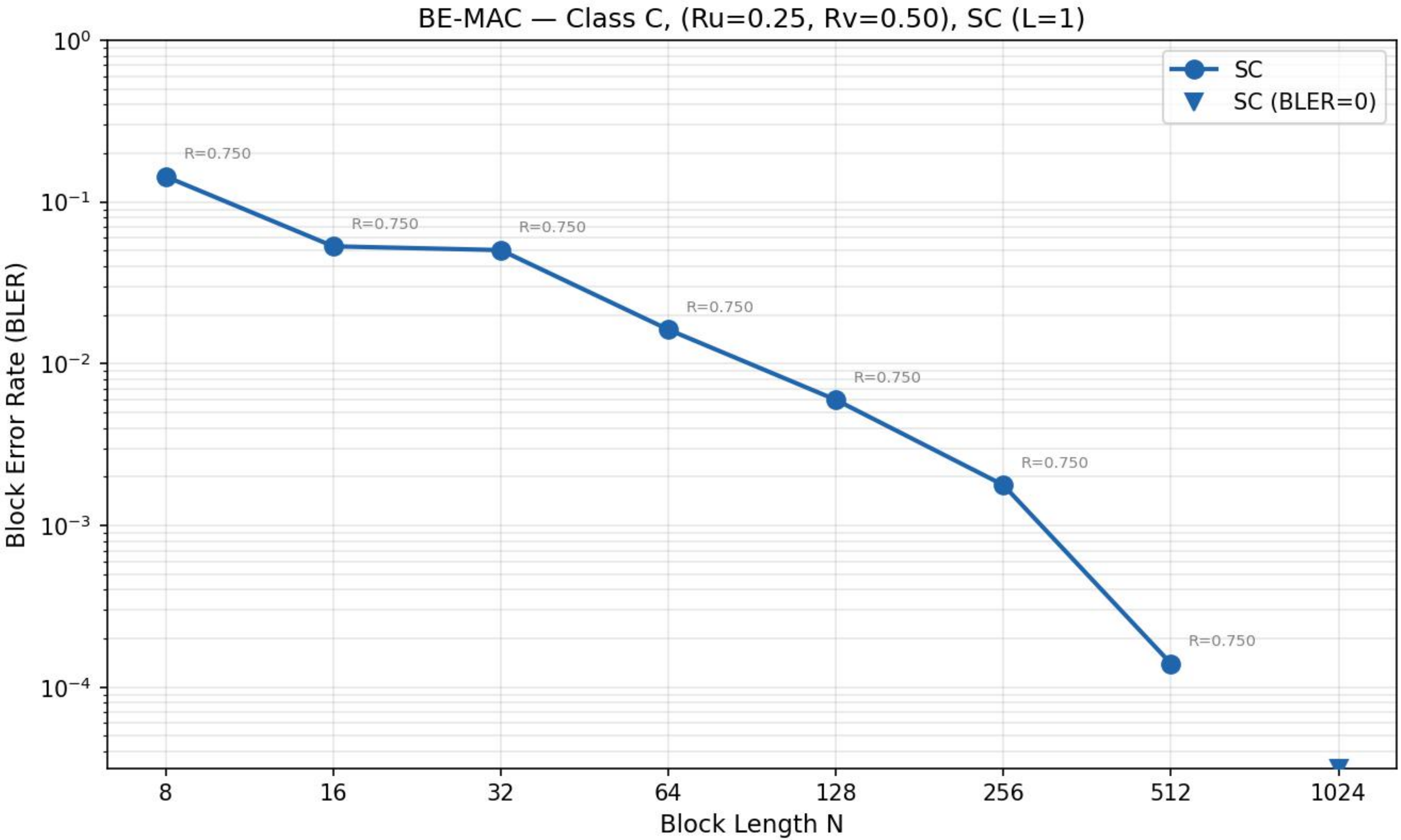
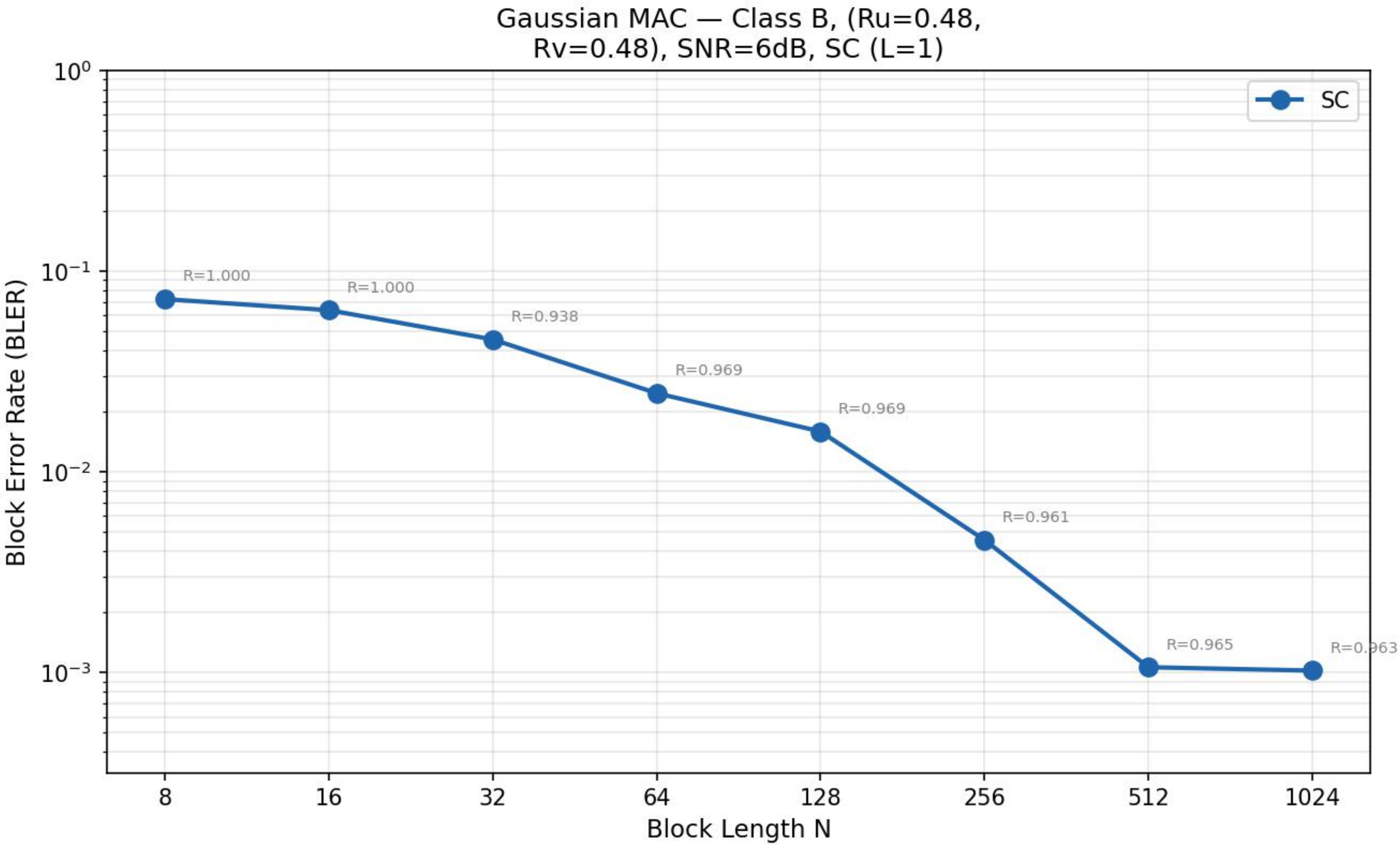
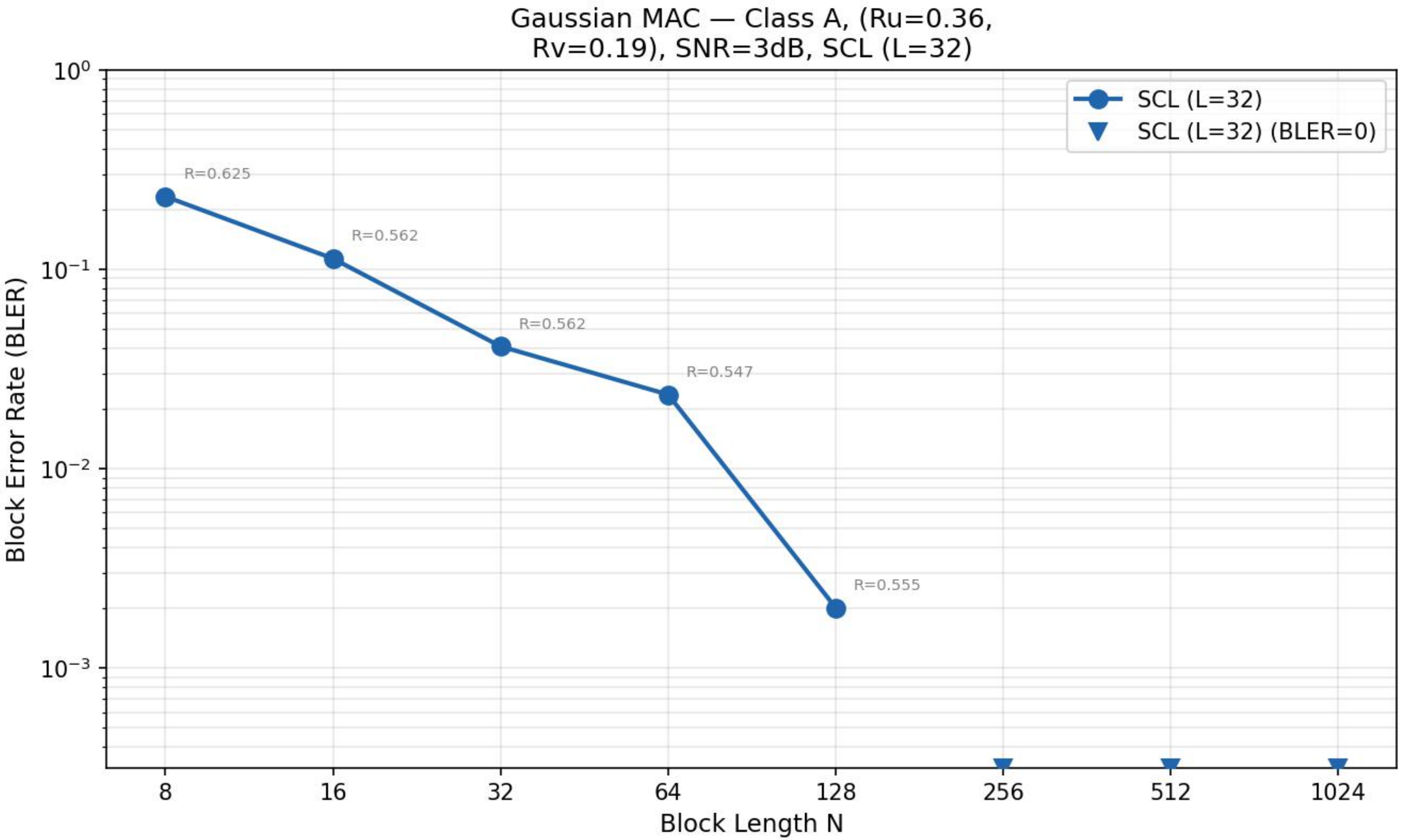
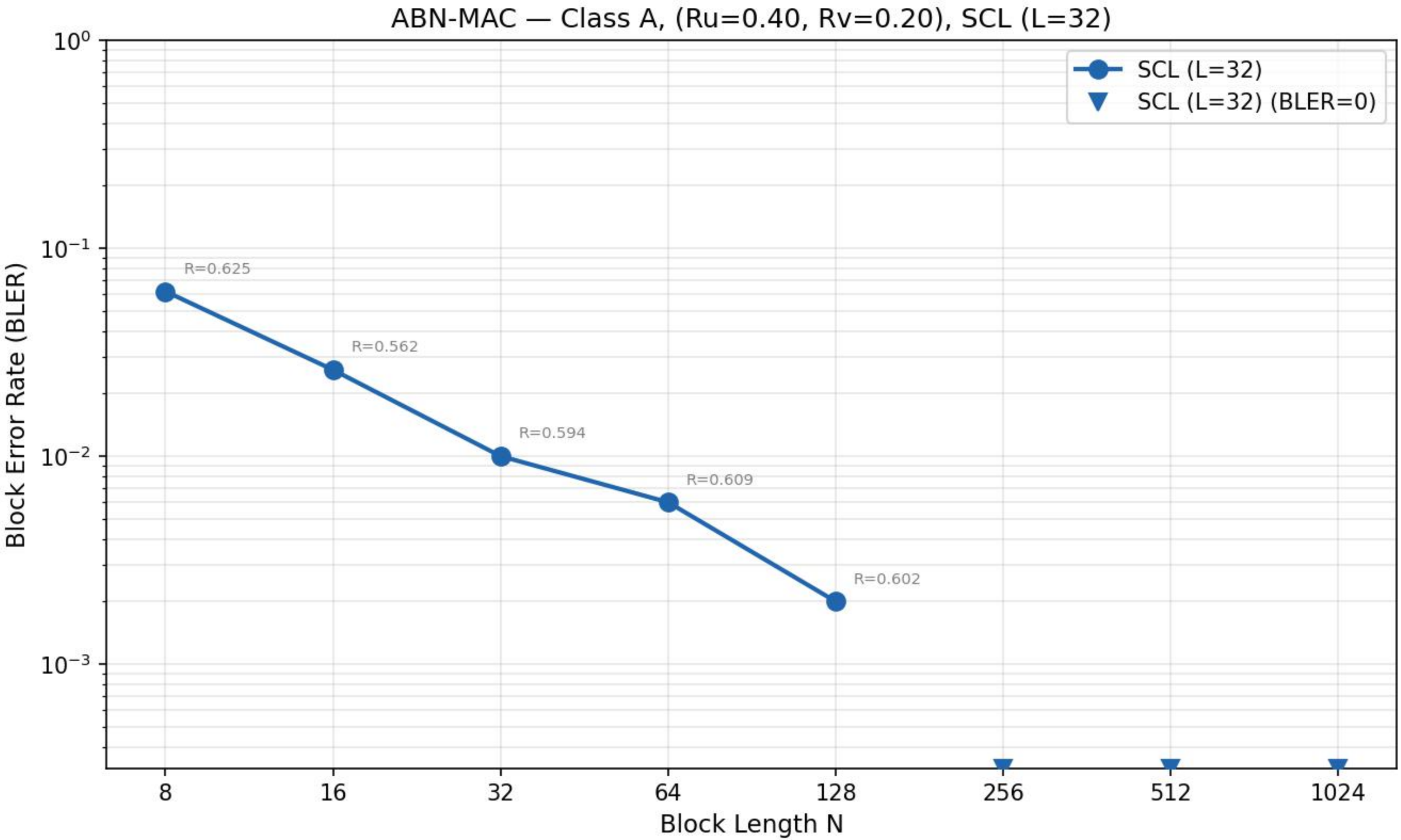
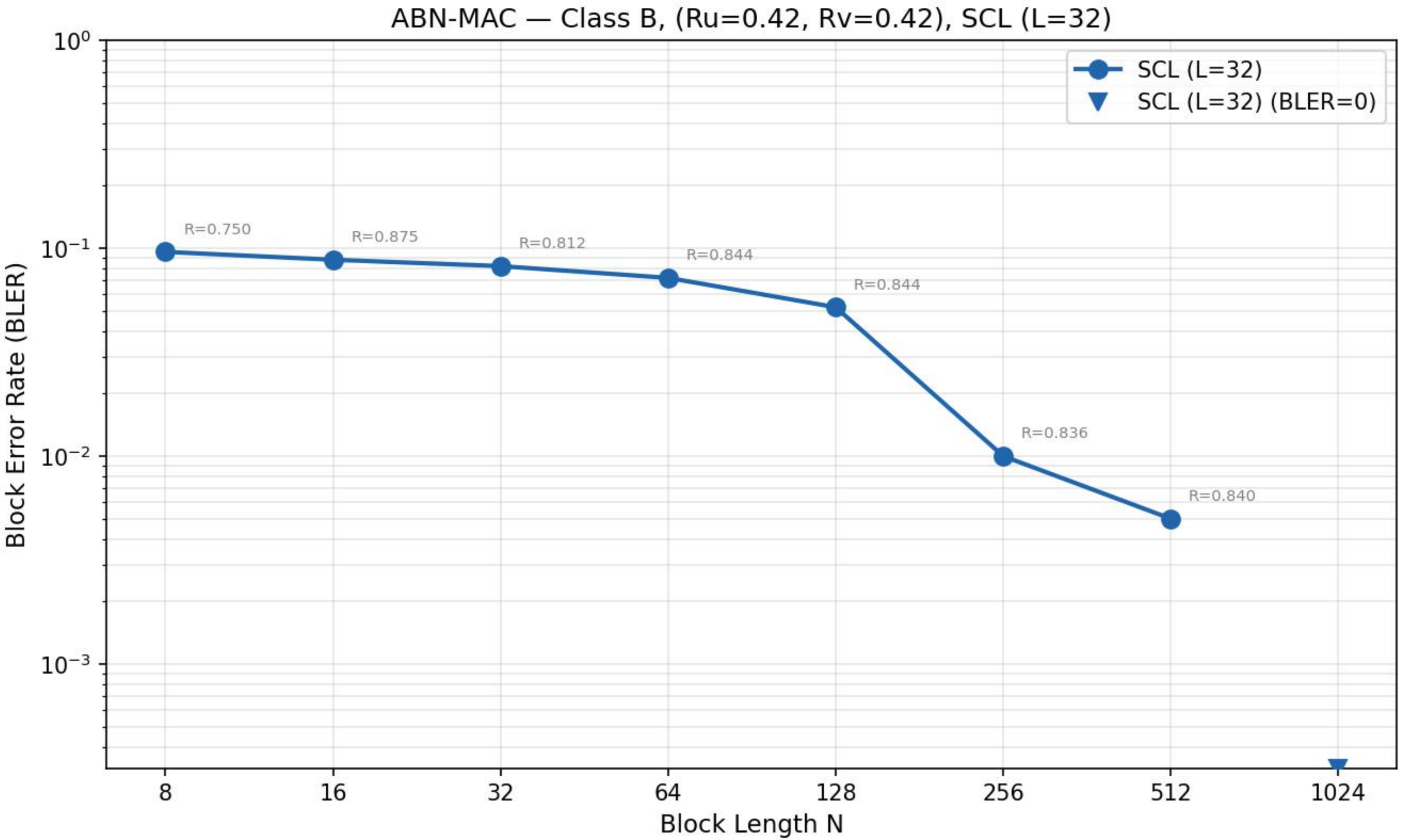
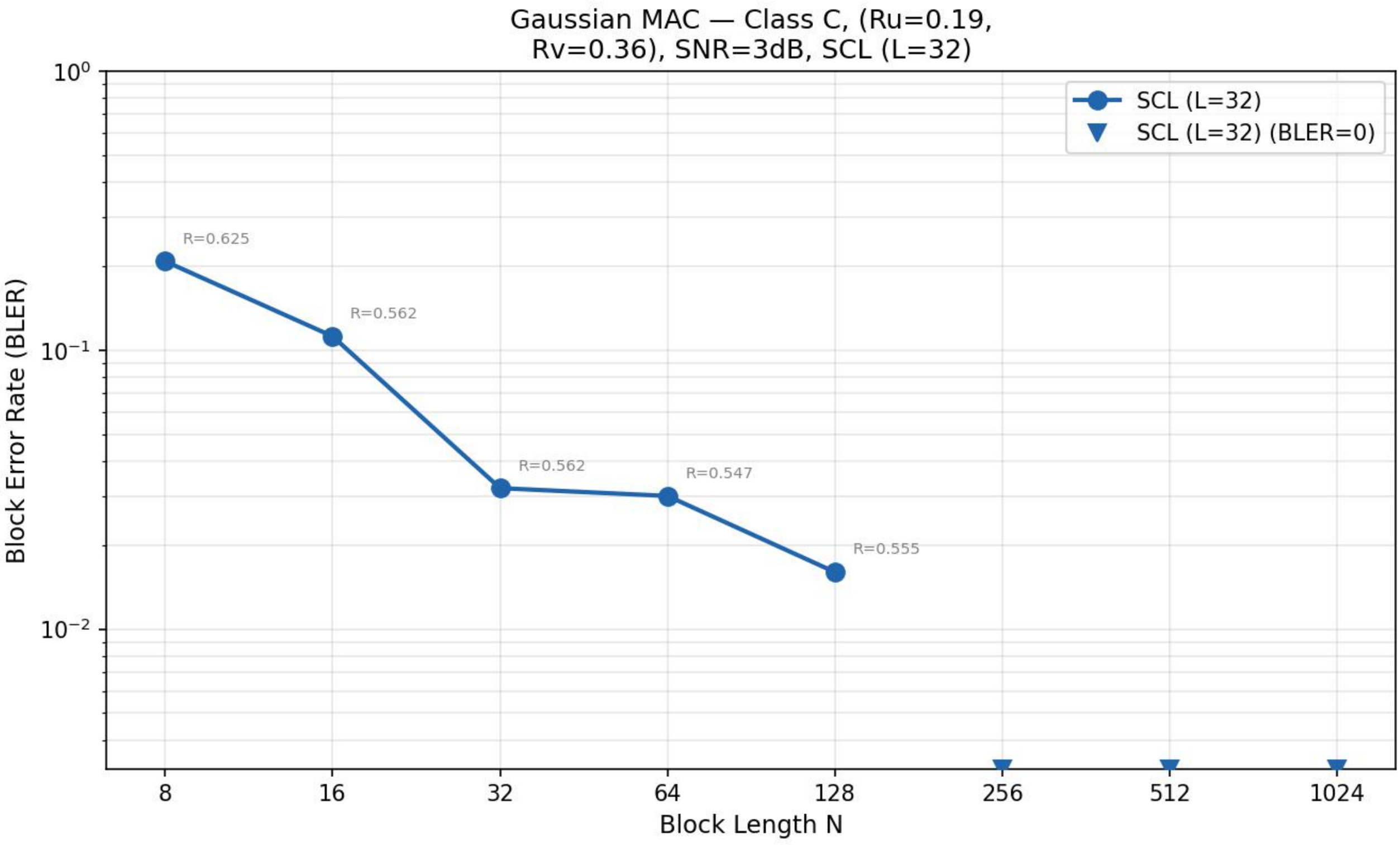

BE-MAC — Class C, (Ru=0.25, Rv=0.50), SC (L=1)
SC
SC (BLER=0)
R=0.750
R=0.750
R=0.750
R=0.750
R=0.750
R=0.750
R=0.750
Block Error Rate (BLER)
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
8
16
32
64
128
256
512
1024
Block Length N

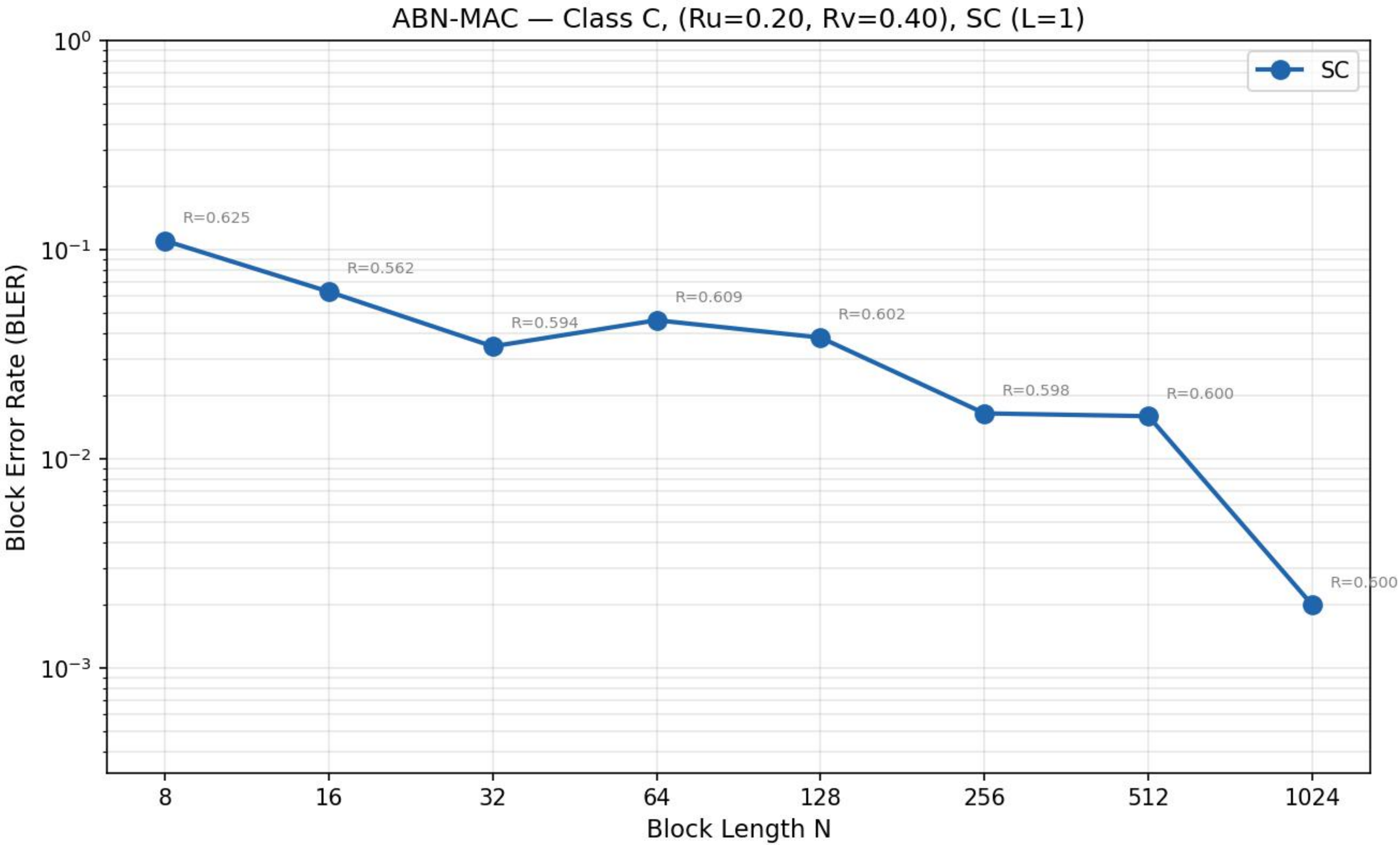

Gaussian MAC — Class B, (Ru=0.48, Rv=0.48), SNR=6dB, SC (L=1)
SC
R=1.000
R=1.000
R=0.938
R=0.969
R=0.969
R=0.961
R=0.965
R=0.963
Block Error Rate (BLER)
Block Length N
10^0
10^{-1}
10^{-2}
10^{-3}
8
16
32
64
128
256
512
1024

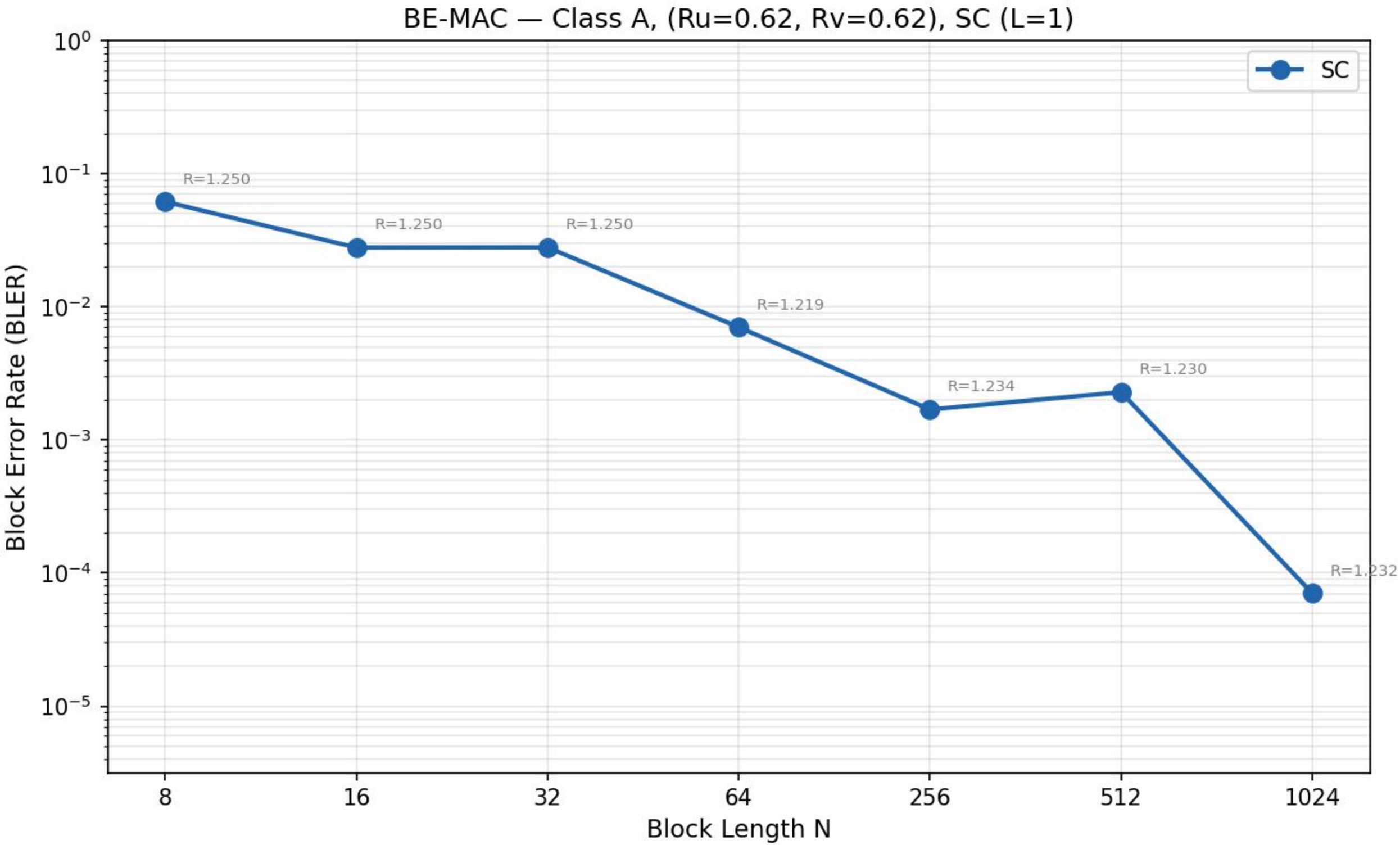

Gaussian MAC — Class A, (Ru=0.36, Rv=0.19), SNR=3dB, SCL (L=32)
SCL (L=32)
SCL (L=32) (BLER=0)
R=0.625
R=0.562
R=0.562
R=0.547
R=0.555
Block Error Rate (BLER)
Block Length N
10^0
10^{-1}
10^{-2}
10^{-3}
8
16
32
64
128
256
512
1024

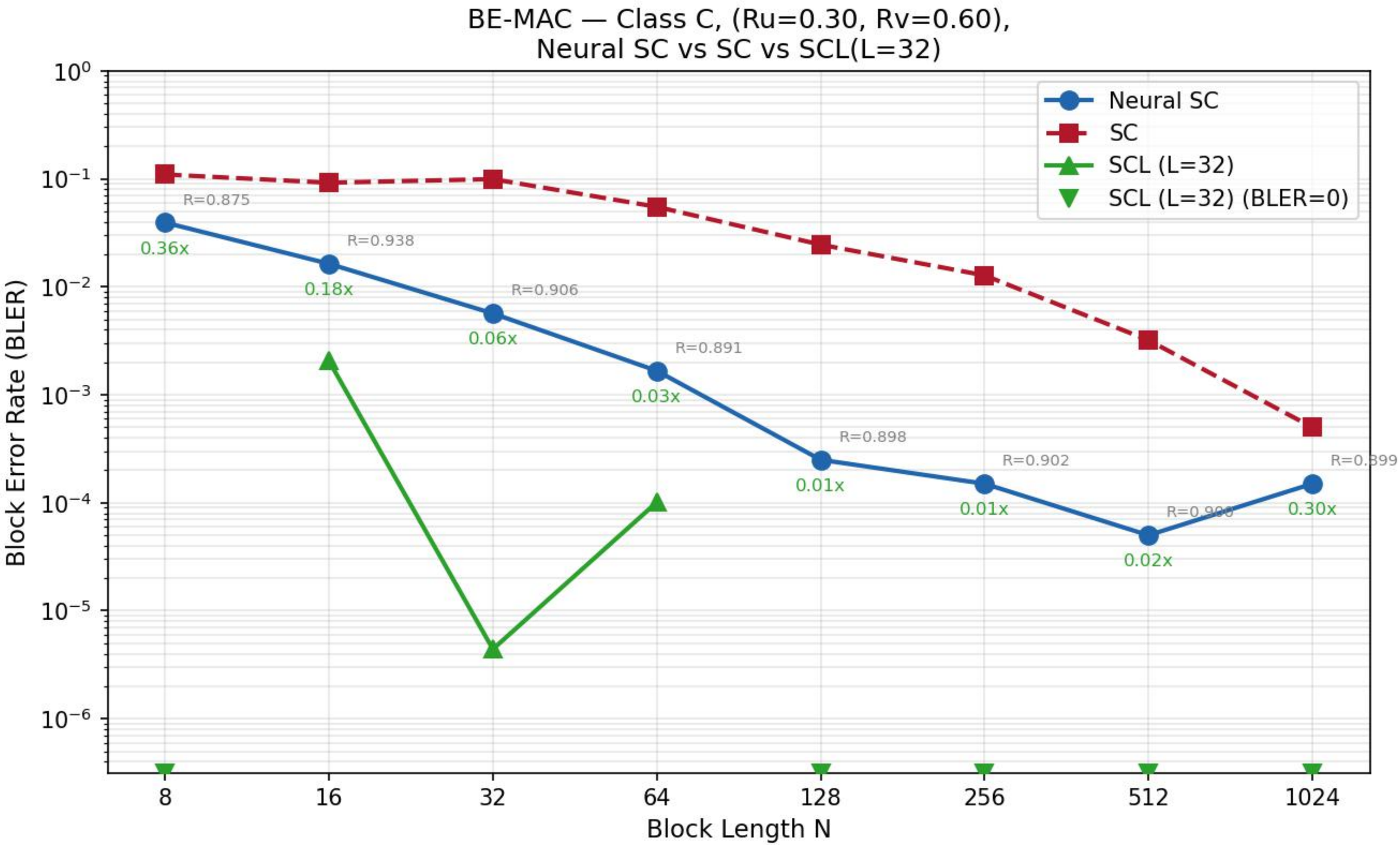

ABN-MAC — Class A, (Ru=0.40, Rv=0.20), SCL (L=32)
SCL (L=32)
SCL (L=32) (BLER=0)
R=0.625
R=0.562
R=0.594
R=0.609
R=0.602
Block Error Rate (BLER)
Block Length N
10^0
10^{-1}
10^{-2}
10^{-3}
8
16
32
64
128
256
512
1024

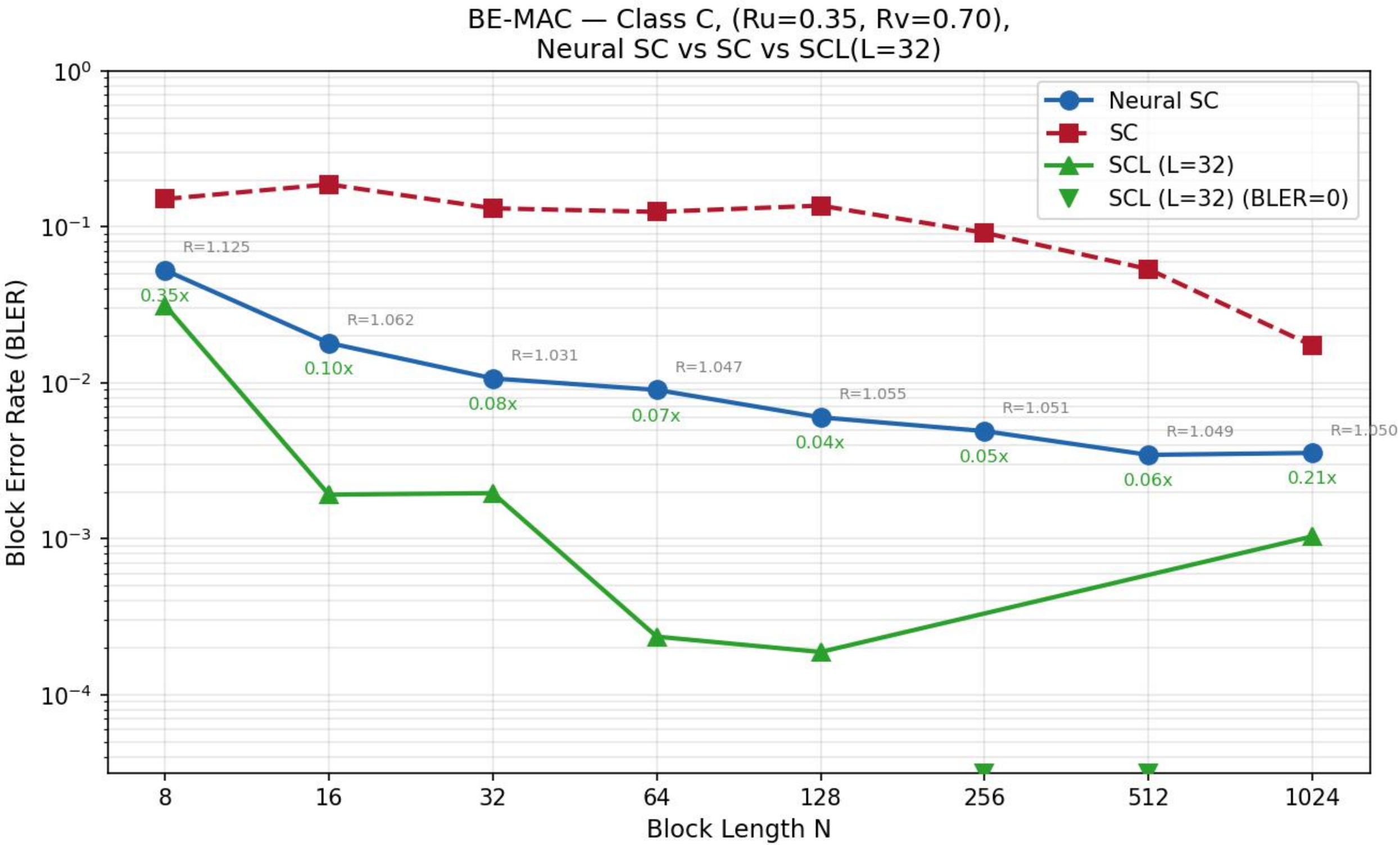

ABN-MAC — Class B, (Ru=0.42, Rv=0.42), SCL (L=32)
SCL (L=32)
SCL (L=32) (BLER=0)
R=0.750
R=0.875
R=0.812
R=0.844
R=0.844
R=0.836
R=0.840
Block Error Rate (BLER)
Block Length N
10^0
10^{-1}
10^{-2}
10^{-3}
8
16
32
64
128
256
512
1024

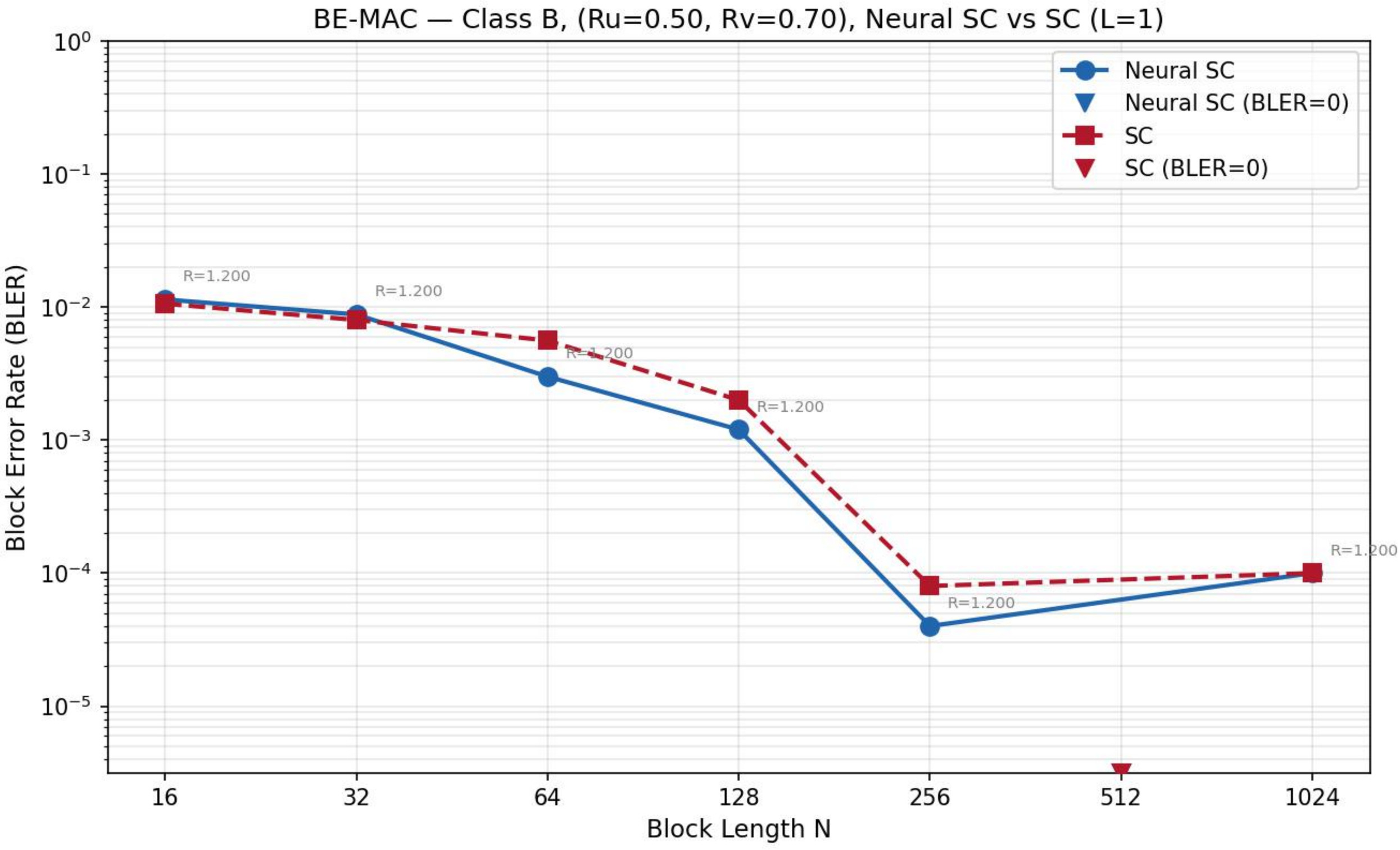

Gaussian MAC — Class C, (Ru=0.19, Rv=0.36), SNR=3dB, SCL (L=32)
SCL (L=32)
SCL (L=32) (BLER=0)
R=0.625
R=0.562
R=0.562
R=0.547
R=0.555
Block Error Rate (BLER)
Block Length N
10^0
10^{-1}
10^{-2}
8
16
32
64
128
256
512
1024

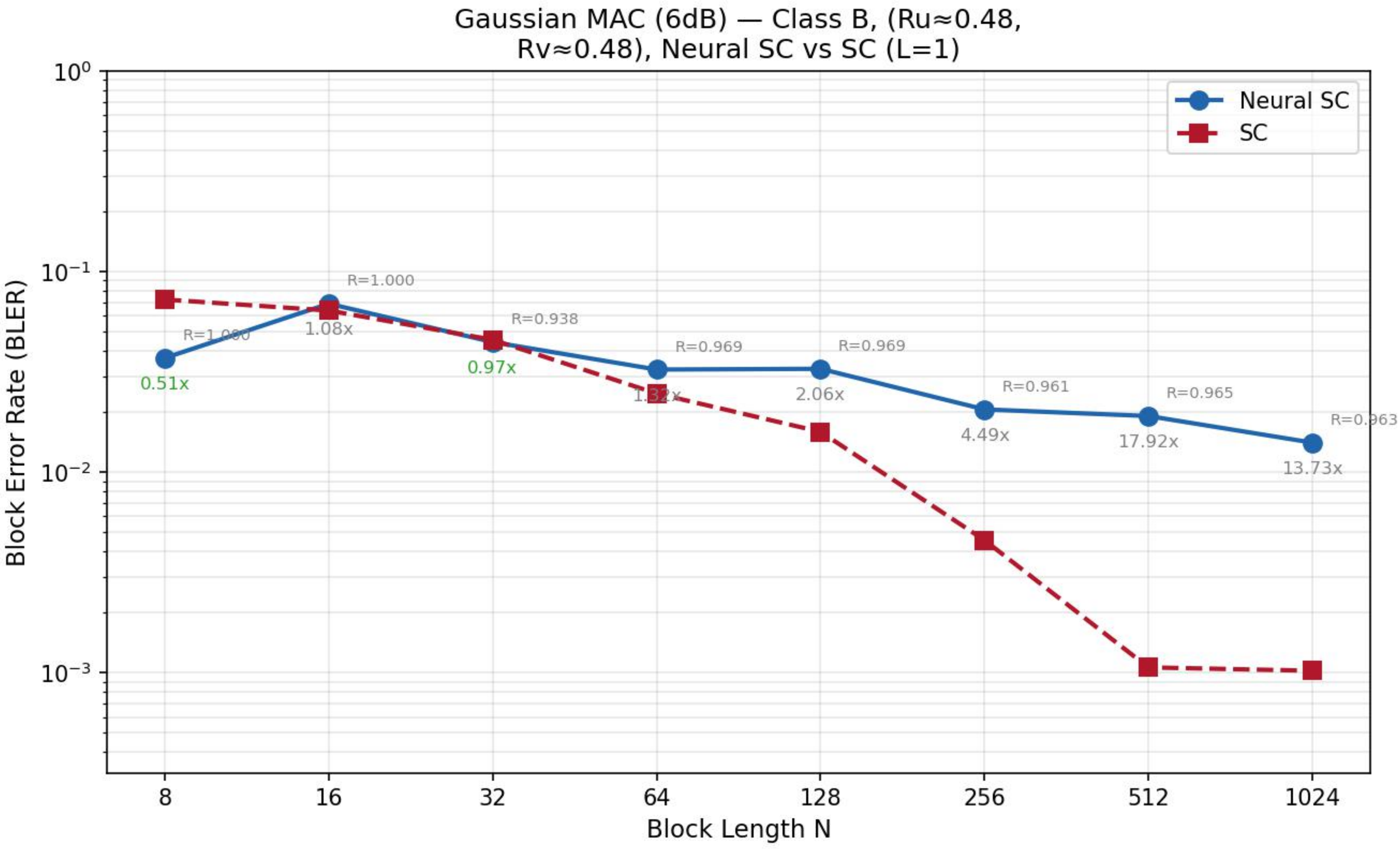

ABN-MAC — Class C, (Ru=0.20, Rv=0.40), SC (L=1)
SC
R=0.625
R=0.562
R=0.594
R=0.609
R=0.602
R=0.598
R=0.600
R=0.500
Block Error Rate (BLER)
10^{0}
10^{-1}
10^{-2}
10^{-3}
8
16
32
64
128
256
512
1024
Block Length N


BE-MAC — Class A, (Ru=0.62, Rv=0.62), SC (L=1)
SC
R=1.250
R=1.250
R=1.250
R=1.219
R=1.234
R=1.230
R=1.232
Block Error Rate (BLER)
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
8
16
32
64
256
512
1024
Block Length N


BE-MAC — Class C, (Ru=0.30, Rv=0.60),
Neural SC vs SC vs SCL(L=32)
Neural SC
SC
SCL (L=32)
SCL (L=32) (BLER=0)
R=0.875
R=0.938
R=0.906
R=0.891
R=0.898
R=0.902
R=0.900
R=0.899
0.36x
0.18x
0.06x
0.03x
0.01x
0.01x
0.02x
0.30x
Block Error Rate (BLER)
Block Length N
8
16
32
64
128
256
512
1024


BE-MAC — Class C, (Ru=0.35, Rv=0.70),
Neural SC vs SC vs SCL(L=32)
Neural SC
SC
SCL (L=32)
SCL (L=32) (BLER=0)
Block Error Rate (BLER)
Block Length N
10^0
10^{-1}
10^{-2}
10^{-3}
10^{-4}
8
16
32
64
128
256
512
1024
R=1.125
R=1.062
R=1.031
R=1.047
R=1.055
R=1.051
R=1.049
R=1.050
0.35x
0.10x
0.08x
0.07x
0.04x
0.05x
0.06x
0.21x


BE-MAC — Class B, (Ru=0.50, Rv=0.70), Neural SC vs SC (L=1)
Neural SC
Neural SC (BLER=0)
SC
SC (BLER=0)
R=1.200
R=1.200
R=1.200
R=1.200
R=1.200
R=1.200
Block Error Rate (BLER)
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
16
32
64
128
256
512
1024
Block Length N


Gaussian MAC (6dB) — Class B, (Ru≈0.48, Rv≈0.48), Neural SC vs SC (L=1)
Neural SC
SC
Block Error Rate (BLER)
Block Length N
10^0
10^{-1}
10^{-2}
10^{-3}
8
16
32
64
128
256
512
1024
R=1.000
R=1.000
R=0.938
R=0.969
R=0.969
R=0.961
R=0.965
R=0.963
0.51x
1.08x
0.97x
1.32x
2.06x
4.49x
17.92x
13.73x